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Time Series Analysis and Forecasting Techniques
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Experiential Learning Activity on a
Customer Churn Rate Forecasting Using Time Series Analysis (ARIMA & SARIMA)

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1. Problem Statement and Significance

1.1 Problem Statement

Customer churn — the phenomenon of subscribers discontinuing a telecom service — represents one of the most costly operational challenges confronting the telecommunications industry. Empirical research consistently demonstrates that acquiring a new customer costs five to seven times more than retaining an existing one. The cumulative revenue loss from churn compounds silently across fiscal quarters as loyal customers are replaced by newer, less stable subscribers. The ability to accurately forecast future churn rates therefore provides a decisive competitive and operational advantage, enabling management to design and deploy targeted retention campaigns well in advance of actual attrition.

This study addresses the following core research question: Can a time-series model, applied to monthly customer churn rates derived from the IBM Telco Customer Churn dataset, reliably forecast churn behaviour 12 months into the future? To answer this, churn rates are aggregated by customer tenure month (t = 0 to 71), producing a 72-point univariate time series. The series is analysed for stationarity using the Augmented Dickey–Fuller (ADF) test, decomposed into trend, seasonal, and residual components via STL, and modelled using ARIMA(0,1,2) and SARIMA(0,1,2)(0,0,0)[12], with model selection guided by the Akaike Information Criterion (AIC).

1.2 Significance of the Study

The significance of this work is appreciated along four complementary dimensions:

Business Impact: Forecasting churn 12 months ahead enables proactive retention campaigns that directly curtail revenue leakage. The IBM Telco dataset exhibits an overall churn rate of approximately 26.5%. Early-warning forecasts allow marketing teams to intervene with personalised offers before at-risk customers act on their intention to leave.
Trend Detection: The STL-linearised trend equation quantifies the well-known but rarely modelled loyalty effect, revealing that each additional month of tenure reduces churn probability by approximately 0.52 percentage points. This has direct implications for customer lifetime value calculations, subscription pricing strategy, and cohort-based marketing segmentationAcademic Rigour: This study demonstrates a complete, reproducible, end-to-end application of the Box–Jenkins ARIMA/SARIMA methodology, covering ADF stationarity testing, ACF/PACF identification, parameter estimation, residual diagnostic checking, and 12-month forecast generation.

Deployment: The entire analytical pipeline is implemented as a React + FastAPI web application, making forecasts interactively accessible to non-technical stakeholders through an intuitive dashboard.

2. Data Collection and Proof

2.1 Dataset Description

The dataset used throughout this study is the IBM Telco Customer Churn dataset, publicly available on Kaggle and IBM's Sample Data repository under the filename WA_Fn-UseC_-Telco-Customer-Churn.csv. The dataset contains records for 7,043 telecommunications customers, each described by 21 attributes spanning demographics, account information, and subscribed services. The target variable is the binary churn label ('Yes' / 'No'), binarised to 1 / 0 for modelling. Customer tenure is recorded in months ranging from 0 to 72, yielding 72 unique tenure groups that form the backbone of the constructed time series. A preliminary data

quality check identified 11 rows with non-numeric values in the TotalCharges column; these were dropped, leaving 7,032 clean records for analysis.

2.2 Time Series Construction

Because the raw dataset is customer-level and cross-sectional in nature, a time series must be explicitly constructed before any forecasting can be performed. This transformation proceeds in three clearly defined steps.

Step 1 — Group by Tenure: For each unique tenure value t, the monthly churn rate is computed as the proportion of churned customers within that cohort: churn_rate[t] = churned_customers(t) / total_customers(t). This collapses the 7,032-row table into a compact 72-row aggregate series indexed by tenure month.

Step 2 — Range Index: The tenure index is replaced by a clean RangeIndex(0, 72) to enforce uniform monthly spacing — a strict prerequisite for ARIMA/SARIMA modelling, as irregular time gaps distort the autocorrelation structure that these models exploit.

Step 3 — Validation: The resulting 72-point series is exported to churn_ts.csv and verified for completeness, with no missing values confirmed. The initial churn rate at month 0 is 61.99%, reflecting high early-tenure attrition typical of telecom subscribers, while the final churn rate at month 71 is only 7.24%, confirming that long-tenured customers are substantially more loyal. The mean churn rate across all 72 observations is approximately 24%. The series is split 80/20 into a training set (months 0–56) and a held-out test set (months 57–71) for out-of-sample model evaluation.

2.3 Key Dataset Statistics

Attribute	Details
Dataset File	WA_Fn-UseC_-Telco-Customer-Churn.csv
Source	IBM Sample Data / Kaggle
Total Records (after cleaning)	7,032 customers
Features	21 (19 predictors + CustomerID + Churn label)
Churn Label	"Yes" / "No" — binarised to 1 / 0
Tenure Range	0 to 72 months (72 unique tenure groups)
Initial Churn Rate (Month 0)	0.6199 (61.99%)
Final Churn Rate (Month 71)	0.0724 (7.24%)
Mean Churn Rate	~0.24 (24%)
Train / Test Split	80% train (months 0–56) / 20% test (months 57–71)

3. Mathematical Background and Statistical Justification

3.1 Stationarity — Augmented Dickey–Fuller (ADF) Test

ARIMA models require the input time series to be weakly stationary — exhibiting a constant mean, constant variance, and an autocovariance structure that depends only on the lag between observations. Stationarity is verified using the Augmented Dickey–Fuller (ADF) test, which tests the null hypothesis H_0 that the series possesses a unit root (i.e., is non-stationary). On the raw series, the ADF statistic was −1.0028 with a p-value of 0.7523, far exceeding the 5% significance threshold — H_0 cannot be rejected, confirming non-stationarity. After applying first differencing, the ADF statistic dropped to −5.8685 with a p-value of 3.28×10^{-7}, well below the 1% critical value of −3.5336, conclusively establishing stationarity. This mandates d = 1 in the ARIMA order selection.

3.2 STL Decomposition

Seasonal and Trend decomposition using Loess (STL) is applied with a period of 12 months to separate the churn series into three additive components:

$$\text{STL Model:} \quad Y(t) = T(t) + S(t) + R(t)$$

where T(t) is the smooth trend component extracted via Loess regression, S(t) captures periodic fluctuations with a 12-month period, and R(t) is the residual noise. The STL results reveal a monotonically declining trend from ≈0.498 at month 0 to 0.052 at month 71. The seasonal component exhibits amplitudes of ±0.09, while residuals fluctuate around zero with no systematic pattern.

The STL trend component T(t) is further linearised by fitting a first-order polynomial via Ordinary Least Squares (OLS) regression over t = 0, 1, …, 71. This yields the key analytical finding of this study:

Key Trend Equation:

$$\text{Trend}(t) = -0.00521 \cdot t + 0.40774 \quad [R^2 \approx 0.96]$$

The negative slope of −0.00521 confirms that the churn rate declines by approximately 0.52 percentage points per tenure month, while the intercept of 0.40774 implies a predicted initial churn rate of ≈40.77% at t = 0. This quantifies the customer loyalty effect with statistical precision: long-tenured customers are dramatically less likely to churn, and this effect is both statistically significant and practically meaningful for business planning.

3.3 ARIMA Model — ARIMA(0, 1, 2)

An AutoRegressive Integrated Moving Average (ARIMA) model is characterised by the order triple (p, d, q), where p is the number of autoregressive lags, d is the degree of differencing, and q is the number of moving-average lags. The general ARIMA model satisfies:

$$\text{ARIMA General Form:} \quad \varphi(B)(1-B)^d y_t = c + \theta(B)\, \varepsilon_t$$

Here B is the backshift operator, φ(B) is the AR polynomial, θ(B) is the MA polynomial, and ε_t is white noise. Parameter selection was performed by pmdarima's auto_arima function via stepwise AIC

minimisation. The optimal model ARIMA(0, 1, 2) was identified: p = 0 indicates no autoregressive terms are required; d = 1 is consistent with the ADF result; q = 2 means two lagged forecast errors capture the residual autocorrelation. The MA(2) process on the differenced series is:

$$\text{MA(2) Process:} \quad (1-B)\, y_t = c + (1 + \theta_1 B + \theta_2 B^2)\, \varepsilon_t$$

3.4 SARIMA Model — SARIMA(0, 1, 2)(0, 0, 0)[12]

SARIMA extends ARIMA by adding seasonal components governed by $(P, D, Q)_s$, where P is the seasonal AR order, D is the seasonal differencing order, Q is the seasonal MA order, and s is the seasonal period. The general SARIMA form is:

$$\text{SARIMA General Form:} \quad \Phi(B^s)\varphi(B)(1-B)^d(1-B^s)^D y = c + \Theta(B^s)\theta(B)\varepsilon_t$$

Running auto_arima with seasonal = True and m = 12 yielded seasonal orders (P = 0, D = 0, Q = 0)[12], indicating no significant seasonal pattern remained after first differencing. SARIMA therefore collapses to ARIMA(0,1,2) for this series, producing identical parameter estimates and performance metrics. This is consistent with the modest seasonal amplitudes in the STL decomposition and confirms that the dominant structure is the long-run declining trend.

3.5 Justification for ARIMA/SARIMA

The choice of ARIMA/SARIMA is justified on five independent grounds: (i) the series is strictly univariate, ruling out multivariate models such as VAR; (ii) the ADF test confirmed a unit root, and ARIMA's differencing operator handles non-stationarity natively; (iii) no exogenous inputs are available, making ARIMAX/SARIMAX unnecessary; (iv) with only 72 observations, deep learning models such as LSTMs are ill-suited due to their data-hungry nature — ARIMA is well known to be robust in the low-data regime; (v) ARIMA parameters carry explicit statistical meaning — MA coefficients, standard errors, and confidence intervals are all directly reportable in an academic context.

3.6 Error Metric Definitions

Model accuracy is evaluated on the held-out test set (months 57–71, n = 15 observations) using three complementary metrics:

$$\text{MAE} = (1/n)\Sigma|y - \hat{y}| \quad \text{RMSE} = \sqrt{[(1/n)\Sigma(y - \hat{y})^2]} \quad \text{MAPE} = (100/n)\Sigma|(y - \hat{y})/y|$$

The Akaike Information Criterion (AIC = 2k − 2 ln L) is used for model selection, penalising complexity to identify the most parsimonious specification.

4. Suitable Tool

4.1 Technology Stack

The project is implemented as a fully integrated full-stack web application spanning data preprocessing, statistical modelling, REST API serving, and interactive frontend visualisation.

Layer	Technology	Purpose
Language	Python 3.11+	Core language for data processing and modelling
Time Series	statsmodels / pmdarima	STL decomposition, SARIMAX fitting, auto_arima AIC selection
Statistics	scipy / numpy / pandas	ADF testing, OLS regression, data manipulation
Classification	scikit-learn	Logistic Regression for individual churn prediction
Backend API	FastAPI	REST endpoints consumed by the React frontend
Frontend	React + TypeScript + Tailwind	Interactive dashboard with charts and prediction form
Charting	Recharts	Time series, decomposition, and forecast visualisations
Static Plots	matplotlib	Residual diagnostics, ACF/PACF plots

4.2 Justification of Tool Choices

Python 3.11+ serves as the sole implementation language owing to its dominance in the data science ecosystem, its extensive library support for statistical modelling, and its seamless integration between data preprocessing, model fitting, and API serving. statsmodels provides a reference-grade SARIMAX implementation that outputs AIC, BIC, parameter standard errors, and 95% confidence intervals directly from the fitted model object, making results fully reproducible and citable. pmdarima's auto_arima automates the Box–Jenkins identification stage — including ACF/PACF analysis, unit root testing, and stepwise AIC minimisation — eliminating subjective trial-and-error from (p, d, q) selection.

FastAPI was selected for its native asynchronous request handling, automatic OpenAPI documentation generation, and high performance for I/O-bound workloads. React with TypeScript enables type-safe component development and real-time interactive visualisation via Recharts. scikit-learn's Logistic Regression pipeline — combining StandardScaler normalisation and OneHotEncoder for categorical variables — provides a complementary individual-level churn predictor alongside the aggregate time-series forecast.

5. Report Solution

5.1 Pipeline Overview

The end-to-end analytical pipeline is organised into four sequential stages. Stage 1: preprocessing.py ingests the raw IBM Telco dataset, performs the group-by-tenure aggregation, computes ADF test statistics, generates ACF/PACF diagnostic plots, and exports the cleaned churn time series to churn_ts.csv. Stage 2: model.py's train_and_export() function fits both ARIMA and SARIMA models using pmdarima and statsmodels, serialises the fitted model objects, and writes forecast values, confidence intervals, and performance metrics to JSON files for frontend consumption. Stage 3: the classification sub-module trains a Logistic Regression pipeline on the full 7,032-row customer-level dataset for real-time individual churn prediction. Stage 4: main.py exposes all outputs through a

FastAPI REST interface with dedicated endpoints for the aggregate time-series forecast, STL decomposition, and the individual customer prediction form.

5.2 STL Decomposition Results

Applying STL decomposition with period = 12 to the 72-point churn series yields three well-separated components. The trend component T(t) traces a monotonic decline from ≈0.498 at month 0 to 0.052 at month 71, reflecting the progressive accumulation of loyal long-tenured customers in the surviving subscriber base. The OLS linearisation of this trend produces:

$$\text{OLS Trend } (R^2 \approx 0.96): \quad \text{Trend}(t) = -0.00521 \cdot t + 0.40774$$

The seasonal component S(t) exhibits modest periodic oscillations of amplitude ±0.09, suggesting mild 12-month cycles possibly linked to annual contract renewal cycles. The residual component R(t) fluctuates around zero across all 72 months with no discernible systematic structure, confirming that the STL decomposition has adequately extracted structured variation.

5.3 Model Selection and Parameters

auto_arima was configured with max p = 5, max q = 5, seasonal = True with m = 12, stepwise = True, and information_criterion ='aic'. The algorithm converged on ARIMA(0, 1, 2) as the globally AIC-optimal specification (AIC = −167.79). The absence of autoregressive terms (p = 0) indicates that past churn rate values add no predictive value beyond what is captured by the two MA lags. Running the same search with seasonal = True confirmed that seasonal orders (P = 0, D = 0, Q = 0)[12] add no improvement.

5.4 Model Performance Metrics

Both ARIMA(0,1,2) and SARIMA(0,1,2)(0,0,0)[12] produce identical metrics on the held-out test set (months 57–71). The MAE of 0.053 and RMSE of 0.062 represent acceptable performance for a parsimonious two-parameter linear model applied to a noisy churn series spanning nearly an order of magnitude (62% down to 7%). The elevated MAPE of 119.18% is a known artefact when actual values approach zero in the later tenure months and is not a reliable indicator for this dataset.

Model	MAE	RMSE	MAPE (%)	AIC	BIC
ARIMA(0,1,2)	0.0530	0.0624	119.18%	−167.79	−161.88
SARIMA(0,1,2)(0,0,0)[12]	0.0530	0.0624	119.18%	−167.79	−161.88

5.5 Forecast Results — 12-Month Horizon

The fitted SARIMA model was used to generate a 12-month ahead forecast covering months 72 through 83, along with 95% prediction intervals. At month 72 the point forecast is 0.1347 (13.47%), with a lower 95% bound of 0.0428 and an upper bound of 0.2266. By month 73 the forecast stabilises at approximately 0.1378 (13.78%), remaining essentially constant through month 83. This stabilisation is a characteristic property of ARIMA(0,1,2): after two forecast steps, the MA terms exhaust their influence and the forecast reverts to a very small downward drift. The 95% confidence intervals widen progressively, reflecting the compounding uncertainty inherent in multi-step ahead ARIMA forecasts.

Month Index	Point Forecast	Lower 95% CI	Upper 95% CI
72	0.1347	0.0428	0.2266
73	0.1378	0.0305	0.2451
74 – 83	0.1378 (stable)	Widening	Widening

5.6 Individual Churn Prediction (Classification Layer)

In addition to the aggregate time-series forecast, the application includes an individual-level churn classifier that predicts in real time whether a specific customer is likely to churn. The classifier is a Logistic Regression model trained on all 19 predictor features from the full 7,032-row Telco dataset. The preprocessing pipeline standardises four numeric features — tenure, MonthlyCharges, TotalCharges, and SeniorCitizen — using StandardScaler, while all 15 categorical features are converted to dummy variables via OneHotEncoder with handle_unknown='ignore' to tolerate unseen categories at inference time. The pipeline is exposed via a FastAPI POST endpoint returning both a binary churn prediction and a probability score, enabling the React frontend to render an interactive real-time churn probability estimator.

6. Conclusions

This study successfully applied ARIMA(0,1,2) and SARIMA(0,1,2)(0,0,0)[12] to forecast monthly customer churn rates from the IBM Telco Customer Churn dataset. The following key conclusions emerged from the analysis.

Declining Churn Trend

STL decomposition and OLS regression confirmed a statistically significant long-term decline in churn rate quantified by $\text{Trend}(t) = -0.00521t + 0.40774$ ($R^2 \approx 0.96$). The slope of -0.00521 translates to approximately 0.52 percentage points of churn reduction per additional month of tenure, providing strong empirical evidence for the customer loyalty effect that underpins subscriber retention economics.

Non-Stationarity and First Differencing

The ADF test confirmed the raw churn series is non-stationary ($p = 0.752$) and that first differencing achieves stationarity at the 1% significance level ($p = 3.28 \times 10^{-7}$). This mandated $d = 1$ in the ARIMA order and is consistent with the presence of a deterministic downward trend in the series.

Optimal Model Specification

auto_arima selected ARIMA(0, 1, 2) as the AIC-optimal specification (AIC = −167.79). The absence of AR terms ($p = 0$) and two MA lags ($q = 2$) indicates that past churn values add no predictive value beyond the two most recent forecast residuals. SARIMA reduced to ARIMA because no statistically significant seasonal structure was detected in the post-differencing ACF/PACF.

Forecast Accuracy

The model achieved MAE = 0.053 and RMSE = 0.062 on the 15-observation held-out test set — acceptable performance for a parsimonious linear model applied to a noisy declining churn series. The elevated MAPE of 119.18% is an artefact of near-zero denominators in later tenure months and should not be interpreted as poor model fit.

Future Churn Stabilisation

The 12-month forecast predicts churn stabilising near 13.78% for months 72–83, consistent with a maturing predominantly long-tenured subscriber base. Progressively widening 95% confidence

intervals underscore the inherent uncertainty of multi-step ahead ARIMA forecasts and suggest that forecast-driven retention decisions should be revisited as new churn data become available.

6.1 Future Scope

Several promising directions exist for extending this work. Incorporating external regressors such as marketing expenditure, pricing changes, and competitor activity via ARIMAX/SARIMAX could explain churn fluctuations driven by exogenous shocks. Facebook's Prophet/NeuralProphet framework offers automatic handling of multiple seasonalities, holiday effects, and structural changepoints, potentially improving performance in the highly non-linear early decline phase (months 0–20). With access to more granular or longer historical data, LSTM and Temporal Fusion Transformer architectures could capture non-linear long-range dependencies beyond the reach of linear ARIMA. Finally, Cox Proportional Hazards survival analysis of individual customer tenure-to-churn would complement the aggregate time-series view by modelling the time-varying hazard of churn at the customer level.

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